Temporal potential structural connectivity, unprotected pinch points, and monitoring priorities for free-ranging cheetahs (*Acinonyx jubatus*) in southern Africa

Liam Crettol

Department of Zoology, Weber State University

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Abstract

[TODO: Context: cheetah range contraction, most range outside protected areas. Needs a source. Weise et al. 2017 for the range itself; (Marker et al., 2008) for the farmland point. Neither is in the .bib yet.]

The southern African Cheetah (*Acinonyx jubatus jubatus*), has had their range decreased drastically in the past decades. Using a multivariate analysis, with the core variables being fixed at the historical baseline, so that the temporal signal is landscape-side. This has only been exacerbated in the past decades due to both natural and anthropogenic reasons, leaving their current extant range majority outside protected areas. This leaves "barriers" like ranchers, roads and other human development in areas where cheetahs will be roaming.

[TODO: Gap: regional work models habitat suitability, not MODELLED change in network permeability. ("observed" is banned here, we never observed permeability.) The 2026 KAZA multispecies assessment (Sousa et al., 2026) excluded cheetah. TRAP: do not imply this is the first cheetah connectivity model. Moqanaki and Cushman 2017 modelled Asiatic cheetah connectivity. Not in the .bib yet.]

Models looking at habitat suitability have been done in the past, but very little predictive modeling of the movement network permeability. One 2026 KAZA (Kavango-Zambezi Transfrontier Conservation Area) multi-species connectivity study excluded the cheetah outright.

[TODO: What we did: held range cores fixed at the 2010–2016 baseline and modelled potential structural connectivity at four snapshots spanning 2012–2024. TRAP: "five predeclared resistance scenarios" is wrong twice. The count is wrong, and the implemented scenarios are not the ones that were predeclared. Say what was actually run or say nothing about the count.] [TODO: Headline result. TRAP: it is not "connectivity loss". Effective resistance FELL slightly on balance, 161 of 253 pairs down, median about -1.2 percent. The honest headline is that permeability barely moved over twelve years, and that the direction is mixed.] [TODO: Second result: protection of the connecting landscape. FACTS: outside the cores, coverage bottoms out near 23 percent at the 80th percentile of current, against a 31.26 percent background, and only rises above background past about the 93rd percentile. TRAP: "share outside protected areas" was the old framing and it came from a comparison that included the fixed cores.] [TODO: Third result: the robust / uncertain split. FACTS: 32 of 45 links robust, 13 kept as survey priorities. TRAP: there are no ranked monitoring CELLS and no alternative weightings. That index was never built. Do not promise it in the abstract.] [TODO: Claim boundary sentence: these are modelled structural priorities, not demonstrated cheetah movement.]

Keywords: landscape connectivity; circuit theory; *Acinonyx jubatus*; protected area coverage; monitoring prioritisation; southern Africa

1 Introduction

The southern African cheetah has had their range substantially reduced, with their current range estimated to be on majority unprotected land. Cheetah range on protected land is governed by centralized park-management initiatives (Buk et al., 2018). Unprotected land poses a different type of challenge, with landowners, local communities and governmental entities involved. This paper looks at the landscape connectivity change between 2012 and 2024, but also provides a predictive look at what future land permeability for cheetah movement may be based on three scenarios: conservation of the land/low-growth, continuation, and high-development. The predictive scenarios are based on the assumption that the core range of cheetahs is fixed at the 2010-2016 baseline, and that any change in permeability is due to landscape changes rather than changes in the core range itself. Which is to say, the core range is not expected to change in the future, but the landscape surrounding it may become more or less permeable to cheetah movement. Based on a multivariate analysis, the results of this study show that the southern African cheetah is at risk of further range contraction and isolation due to anthropogenic development, and that conservation efforts should focus on maintaining and enhancing connectivity between core range areas.

Move 2: What is already modelled, and what is not

Previous research in the region has predominantly focused on habitat suitability, a topic that is already comprehensively addressed in the existing literature. The objective of my study is not to develop an additional suitability model, but rather to identify and evaluate spatial “pinch points” (i.e., constricted linkages) among areas of suitable habitat that occur on unprotected lands. Importantly, habitat suitability does not necessarily correspond to land that is functionally usable by the species its identified for. Therefore, assessing connectivity among usable habitat is a key gap that this study seeks to address. To my knowledge, no published studies have quantified how landscape permeability through these core areas has *changed* over time. The study most comparable to the present work is a recent multi-species connectivity assessment; however, that analysis explicitly excluded cheetah due to insufficient data. Data limitations were therefore explicitly considered and mitigated to the extent possible in this study, and they provide both the primary justification for the existence of this paper and its principal constraint.

[TODO: Check the novelty claim before this goes out. "No published studies have quantified how landscape permeability through these core areas has changed" is defensible for THIS system, but connectivity modelling for cheetah is not new. Moqanaki and Cushman 2017 modelled Asiatic cheetah connectivity in Iran, and it is not in the .bib yet. Narrowing the claim to southern Africa and to temporal change costs you nothing and is simply more accurate. Also: the multi-species assessment is (Sousa et al., 2026), worth citing by name rather than "a recent multi-species connectivity assessment".]

Move 3: The methodological frame

Two primary connectivity approaches are commonly used to model these kinds of frameworks: least-cost connectivity and circuit-theoretic connectivity (Adriaensen et al., 2003; McRae et al., 2008). The least-cost approach links core areas by selecting the route with the lowest cumulative “cost,” meaning the path of least resistance between cores, based on weight values assigned to each multivariate resistor. In contrast, the circuit-theoretic approach represents the landscape as a resistor network, sends a “current” through it, and highlights redundancy across the broader landscape, relying on the same multivariate resistors. A key limitation of both methods is that you must first assign a resistance “score” or “weight” to every cell (1 km in my case). Zeller et al. (2012) demonstrated that even reasonable alternative assumptions about resistance values can influence results more strongly than the choice of connectivity method itself. For that reason, rather than treating one resistance surface as definitive, I varied the weights along two axes. The first changed the balance among the anthropogenic pressures, comparing my primary weighting against an equal weighting and an infrastructure-emphasis weighting. The second changed how much the surface leaned on vegetation relative to anthropogenic pressure and terrain, at 10, 20 and 40 percent. The vegetation-balanced surface combined with the documented fence treatment carries the primary results, and the remaining combinations are reported as sensitivity rather than as alternative answers. A link is only described as robust here if it survives them. This helps to minimize bias from the selection of resistor weights.

[TODO: Sentence above now matches what was actually run. Two small things left: 1. The protocol still calls these RES-01 to RES-06. Worth a line in the protocol saying the implemented set differs, mostly so your own record stays straight. 2. Cites worth adding here: Zeller et al. (2018) alongside Zeller et al. (2012), and Beier et al. (2008) for the design-choices point.]

Move 4: Objectives and hypotheses

The first objective was to measure changes in modeled structural connectivity among fixed-range cores across four time points from 2012–2024. The second was to determine pinch points that remain influential across the entire sensitivity grid and describe their protected-area status. The final objective was to produce and stress-test a ranked list of monitoring priorities. I made the following not mutually exclusive predictions in support of each hypothesis.

H1 (revised from the workbook). Connectivity loss is concentrated on a minority of network links rather than distributed across the network, and the affected links are disproportionately those with high betweenness in the baseline network.

Note on the revision. H1 as originally drafted predicted that cells with larger increases in human pressure would show larger increases in resistance. That is true by construction, since the pressure layers are the resistance inputs. It cannot fail and therefore is not a hypothesis. The revised form asks a question the model can answer negatively: pressure change might fall on redundant links, in which case network-level connectivity is robust to it and the conservation implication reverses.

Refuted if: connectivity loss is approximately uniform across links, or loss falls preferentially on low-betweenness links.

H2. High-value connectivity area is disproportionately located outside formal protected areas.

Refuted if: the outside-PA share of high-current area does not exceed the predeclared null (§2.9) at the stated significance level, or the result does not hold across current-flow thresholds.

H3. A limited subset of transboundary linkages remains important across all resistance assumptions, while other apparent corridors are strongly model-sensitive.

Refuted if: agreement is uniformly high (in which case the ensemble adds nothing) or uniformly low (in which case no robust set exists and the paper's primary output becomes the uncertainty map, per the stopping rule in the predeclared analysis protocol).

H4. The highest-priority monitoring cells occur where structural importance coincides with old or sparse cheetah evidence and rapid landscape change.

Refuted if: no stable top tier persists across the predeclared weight sets.

[TODO: Clause above is taken from the protocol, so H4 now matches H1 to H3 in form. The real issue is that you never built the weight sets or the cell-level index, so there is nothing to test stability across. Simplest thing: keep H4 as written and say in Results that you did not get to it, and why. That is a normal outcome for a project this size and it reads better than a result you had to stretch for. If you would rather restate H4 around the link-level robust / uncertain split, that is fine too. Just note somewhere that you changed it and roughly when, so the decision trail stays legible. Do not just delete it. HOW EACH ONE ACTUALLY LANDED, so the wording here matches Results: H1: not supported. Change was small and spread across the ranking, not concentrated on high-betweenness links. H2: supported for the broad connecting landscape, but it reverses above about the 93rd percentile of current. The null it refers to changed, so the refutation clause needs a small reword. H3: partly. 32 of 45 links is a majority, not the "limited subset" predicted. H4: no result at all. The monitoring index was never built. ONE SMALL FIX LEFT: the third objective promises "a ranked list of monitoring priorities", which does not exist.]

Terminology. Throughout, potential structural connectivity refers to modelled landscape permeability between fixed range cores. It is not evidence of cheetah movement, dispersal, or gene flow. Where a location is described as a pinch point, this means a cell of high modelled current concentration, not an observed crossing.

2 Methods

2.1 Study area and analysis extent

The study area was defined as the southern African cheetah range [TODO: Weise et al. 2017, PeerJ 5:e4096. Not in the .bib yet.] plus a buffered analysis extent. All layers were aligned to a fixed equal-area projection and a 1 km snap grid, so every raster is directly comparable cell for cell. Cores were derived from the historical confirmed-range baseline and held constant across 2012, 2016, 2020, and 2024. The final network consisted of 23 cores and 45 selected neighboring core pairs.

2.2 Rationale for the working resolution

I used a 1-km working resolution because it represents a compromise between the rougher 10-km occurrence baseline and the finer environmental covariates, which range from 30 m to 1 km. This resolution is fine enough to identify meaningful points in the landscape while avoiding the implication of a level of accuracy that the occurrence evidence alone could not provide.

2.3 Range cores

I defined a core area as a contiguous patch of cells that were confirmed presence, with a density of ≥ 0.51 cheetahs per 100 km^2. The final network consisted of 23 cores, which were held constant across all four temporal snapshots to avoid inventing core movement in the absence of comparable range reassessment.

I represented these 23 core areas as fixed nodes across all four temporal snapshots (2012, 2016, 2020 and 2024). I took each core area and calculated the Euclidean distance to the nearest neighbor. This yielded 45 straight-line links. These straight-line links defined the lines used to get the least cost paths based on the added resistance surface rasters.

2.4 Covariates and their temporal treatment

The predicted temporal resistance values are coming from the continuous rasters (built-up fraction (human development) from GHSL, nighttime lights from VIIRS, and VCF (vegetation) from MODIS). Categorical land cover is used for masking and description only, because annual categorical features switch classes between years for reasons unrelated to overall landscape change.

I also used the static covariates (roads, livestock, terrain, hydrography, fences, protected areas, and core definitions). The static covariates do not contribute *temporal* signal to the resistance surface, however they do contribute *spatial* signal to the resistance surface. The static covariates are used to describe the landscape and to provide context for the dynamic covariates.

2.5 Resistance scenarios

[TODO: WF-013, WF-014. SAY: what the final resistance surface is made of, and that the scenario set was fixed before any result was looked at. FACTS: final model = anthropogenic 0.60, vegetation 0.20, terrain 0.20, then multiplied by a fence multiplier running 1.0 to 25.0. Inside the anthropogenic 0.60: built 30, roads 30, livestock 30, lights 10. Sensitivity scenarios = vegetation low (0.7/0.1/0.2), balanced (0.6/0.2/0.2), high (0.4/0.4/0.2). TRAP: RES-01 to RES-06 never existed as such. Say what actually ran. five in main.tex says five and needs fixing. CITE: Zeller et al. (2012) on transformation mattering as much as weights.]

[OUTLINE]

Terrain component. The DEM was not used directly as resistance; slope was derived from it. Mean slope carries the primary terrain surface, maximum slope retained for sensitivity and context. Terrain resistance function: slope $\leq 10°$ low resistance; 10 to 30° linearly increasing; $\geq 30°$ capped high; resistance rescaled to 1 to 10.

Original primary temporal surface. Anthropogenic component 80% of total, terrain 20%. Within the anthropogenic component: built-up 30%, roads 30%, livestock 30%, nighttime lights 10%.

Three internal weighting models compared: primary, equal, infrastructure-emphasis.

Vegetation correction. The first temporal model omitted VCF from the resistance calculation, which is a mismatch with the predeclared protocol. Three further sensitivity scenarios were added, keeping the original anthropogenic-only surfaces intact rather than replacing them:

- vegetation-low: 10% vegetation, 70% anthropogenic, 20% terrain
- vegetation-balanced: 20% vegetation, 60% anthropogenic, 20% terrain
- vegetation-high: 40% vegetation, 40% anthropogenic, 20% terrain

The vegetation proxy uses continuous VCF fields to represent sparse and bare cover. It does not claim to measure fine-scale habitat edge or observed movement preference, and should be described that way.

Reconciliation still needed: these six weighting models are not the same object as the predeclared RES-01 to RES-06 scenario set. Either map each one onto a predeclared scenario ID or state plainly in the text that the implemented set differs and why.

Resistance values were assigned to represent the relative difficulty of movement through each landscape feature. The terrain component was derived from slope, with steeper slopes assigned higher resistance values. This had to be done carefully however, given cheetahs do have an affinity for some sloped terrains [TODO: Cite the preffered terrain source here]. I compared alternative vegetation weights because vegetation cover can significantly influence cheetah movement, and different weighting schemes allow for sensitivity analysis of this factor.

2.6 Connectivity modelling

[TODO: WF-015, WF-016. SAY: how paths and current were computed, and why every current result is reported outside the cores. FACTS: least-cost = one CostDistance per source core, then CostPathAsPolyline per destination (equivalent to pairwise runs, say why). Current = Circuitscape.jl 5.17.1 (Anantharaman et al., 2020), pairwise, 253 pairs per year, four years, 1012 solves, CG+AMG, double precision, eight-neighbour. Finished 8 Sept 2026, no failures. Graph has 34 connected components. THE IMPORTANT ONE: in pairwise mode every core is a source or a ground, so the cores themselves carry the highest current. The entire top decile sits inside them. That is why all current results use the outside-core domain. The Circuitscape 5 setting that would switch this off is not implemented, so we mask afterwards rather than re-solve. Say that plainly, including that masking cleans the numbers but not the solve. ALSO: why not omnidirectional (Landau et al., 2021). Cores are defined in advance, so core-to-core is the right question.]

Corridor definition. [TODO: Already decided, just write it. SAY: corridors are continuous current surfaces summarised by percentile. Thresholded polygons are for maps only, never for a statistic. Percentiles are taken on the outside-core domain (see §2.6), and the bands were fixed before the coverage result was looked at.]

2.7 Network structure

[TODO: WF-017. SAY: we used one network metric, betweenness, and dropped the rest on purpose. DROPPED: PC and *dPC* (Saura & Pascual-Hortal, 2007), Conefor, and the dispersal-distance parameter. Reason to give: PC needs a dispersal kernel, and no cheetah dispersal telemetry exists for this system, so the number would look more precise than it is. Say it was dropped. Do not just leave it out. WHAT WE HAVE: edge betweenness on the 45-edge graph, unweighted, normalised by the 253 pairs. Admit it is unweighted, because the protocol said effective distances would be the weights. THE IMPORTANT ONE: the ranking depends on how the link set was built. Links = union of a 3-nearest-neighbour graph and the minimum spanning tree. At k = 2/3/4/5 you get 31/45/61/77 edges. Only k = 3 has a bridge (17-24), and the top-ranked links change at k = 4 and 5. So report a rank range across k, not one ranking. CITE: Keeley et al. (2021) on choosing metrics to fit the question.]

[OUTLINE]

edge betweenness computed on the 23-core network, link importance ranked on normalised edge betweenness. PC and *dPC* not yet computed, and dispersal distance is still the 100 km placeholder.

2.8 Temporal comparison

[TODO: WF-018. Same cores, same scenarios, same resolution across all four snapshots, so that differences are attributable to covariate change alone. Give the change equations. Report both absolute and proportional change.]

2.9 Protected-area coverage

[TODO: WF-019, CAU-011, CAU-012. Rewrite this one from scratch. WHY THE OLD PLAN IS DEAD: full extent versus modelled network was never the real choice. Both included core cells, and the cores are fixed, so both measured something that cannot change. SAY: how well protected the connecting landscape is, cores excluded. FACTS: WDPA <MONTH YEAR> release, August 2026. Polygons only, terrestrial and coastal, Proposed excluded, dissolved, 2301 in extent. Point records and marine-only polygons dropped. Background = 31.26 percent. Coverage is lowest at the 80th percentile of current, about 23 percent. It crosses background near the 93rd. It reaches about 45 percent at the 99th. The gap widened between 2012 and 2024. SAY WHY 31.26 AND NOT 30.13: the wider figure includes 165,765 zero-current cells that could never have been selected. TRAP: do not quote a p-value. It moves from 0.006 to 0.096 just by changing the assumed spatial block size. Lean on direction holding across all four years and across 0, 1, 5 and 10 km core buffers instead.]

2.10 Robustness and uncertainty

[TODO: WF-022. NOT DONE: the planned 144-run grid. One core definition, 1 km only, no 2 km or 5 km check, no cell-level agreement map. Say so, and move the resolution claim in §2.2 down into Limitations since nothing tested it. SAY: what "robust" actually meant here. It is a link-level screen, three gates, all must pass. 1. Weight. Both alternative weightings keep 80 percent of the route within 5 km of the primary route. 34 of 45 pass. 2. Fence. Five links move (1-13, 1-17, 9-13, 13-17, 27-30). Leaves 32. 3. Vegetation direction. Change must keep the same sign under all three vegetation weights. Two fail (28-38, 40-45), both already out. Final: 32 primary, 13 uncertainty. TRAP: that 80 percent is route overlap. It is NOT the predeclared cell-level k/n = 0.8 agreement rule. Same number, different thing. Say so.

SECOND POINT WORTH MAKING: cost is stable, location is not. 49 of 180 route-years fell below 80 percent overlap at 1 km, covering 21 of the 32 "robust" links, and 17-24 hit zero in 2020. Report those as two separate things. Note the two overlap tests use different distances (5 km and 1 km) and are not one criterion.]

I considered a res

[OUTLINE]

- 90 weight comparisons evaluated.
- 34 of 45 core pairs robust under the tested alternative weights, 11 sensitive.
- Fence scenarios changed 5 links: 1–13, 1–17, 9–13, 13–17, 27–30.
- 32 links classified primary-priority eligible; 13 retained as uncertainty and data-collection priorities.

Two things to resolve before writing: state the numeric robustness rule actually applied and whether it matches the predeclared $k/n = 0.8$; and reconcile the 34 robust pairs with the 32 primary-priority-eligible links, since those are different counts and a reader will notice.

2.11 Independent plausibility assessment

[TODO: WF-020, WF-021, CAU-004, CAU-017. "Validation" stays a banned word here. THE PROBLEM: the independent check we predeclared, the Limpopo tracking data and the Namibian camera-trap work, was never run. What exists is a comparison against 2010-2016 occurrence records from the same archive that built the cores, so it cannot be independent. DECIDE: run the Limpopo check, or rename this subsection and say the independent check was not done. Either is fine. Just do not let the existing screen read as though it were the independent one, because it is built from the same records as the cores. IF IT STAYS: early 2010-2012 = 413 thinned 10 km cells, late 2013-2016 = 450, Jaccard 0.239. And say there are no records after 2016, so 2020 and 2024 have no check at all. Roberts et al. (2017) only matters if the empirical-resistance route comes back.]

2.12 Monitoring prioritisation

[TODO: WF-023, WF-024, CAU-020. NOT BUILT: the six-component MCDA at 10 km with weight sets W1-W3. SUGGESTION: a link-level decision table instead. A weighted composite would imply you can trade these things off against each other, and there is no basis for the exchange rates. SAY: the output is a table with the fields kept separate, not scored together. Structural importance (betweenness, with its k-sensitivity), 2012-2024 cost direction and size, route-location stability, fence uncertainty, protected-area context. State that no composite is computed and that a reader can weight them however they like. Rank stability, if reported, is across k, not across weight sets.]

[OUTLINE]

link ranking currently rests on normalised edge betweenness plus the robust / sensitive split, with 32 primary-priority-eligible and 13 uncertainty-priority links. The full MCDA with three weight sets is not yet built.

2.13 Software and reproducibility

[TODO: KNOWN: Circuitscape.jl 5.17.1. STILL NEEDED: ArcGIS Pro [TODO: version], Julia [TODO: version], Python [TODO: version]. Google Earth Engine for SRTM, GHSL, VIIRS, MODIS MCD12Q1 and VCF. Most geoprocessing is arcpy, not ModelBuilder. The reanalysis in resistances/ uses GDAL instead; mention that if it gets cited. Archive location, and what can and cannot be redistributed (CAU-012, CAU-013).]

[OUTLINE]

ArcGIS Pro Spatial Analyst via arcpy (`CostDistance`, `CostPathAsPolyline`); Google Earth Engine for SRTM, GHSL, VIIRS, MODIS MCD12Q1 and VCF acquisition and harmonisation. Record versions.

3 Results

3.1 Connectivity change, 2012–2024

[TODO: H1. Magnitude and geography of change. PRIMARY MEASURE: effective resistance. Least-cost path cost is secondary and gives bigger numbers for the same links; say which is which. FACTS: 253 pairs, median -1.224 percent, median absolute 2.822, 90th percentile 6.089, 92 up and 161 down. The 45 selected links: median -1.557. The 32 priority links: median -2.618, 12 up and 20 down. Biggest single move +13.1 percent on 33-38. Outside-core current: median relative change 4.3 percent, top-decile agreement between years 0.91 to 0.93. TRAP: "report per country" cannot be done. Cores have no country attribution and the only per-country table is covariate change, not connectivity. Either build the attribution or drop the promise. Figure 1.]

3.2 Robust priority links

[TODO: H3. How many links met the criterion and how concentrated they are. FACTS: 32 of 45 primary, 13 uncertainty. Weight screen passes 34, fence screen removes 2 more. Vegetation-direction removes none that were still in. NOTE: H3 predicted a LIMITED subset. 32 of 45 is most of the network, so it is only partly supported. Worth saying, since it is a more interesting result than a clean confirmation. Figure 2, Table 2.]

3.3 Protected-area coverage of priority connectivity

[TODO: H2. Protection of the connecting landscape. Rewrite, do not fill in. WHY: the predeclared comparison included the fixed cores, and the whole top decile of current sits inside them, so it measured a footprint that cannot change. FACTS, outside cores, background 31.26 percent: 80th percentile 23.27, 90th 25.54, 95th 33.45, 99th 44.63. Crossover near the 93rd. Gap widened from -7.03 to -7.99 points at the 80th between 2012 and 2024. Direction holds at 0, 1, 5 and 10 km core buffers. TRAP: no p-value. It swings 0.006 to 0.096 on block size alone. Also drop the word "observed"; this is modelled current. Table 3.]

3.4 Model sensitivity

[TODO: Model sensitivity. Still substantial, but not the subsection that was planned. DOES NOT EXIST: the cell-level agreement grid, and any agreement raster or map. WHAT TO REPORT INSTEAD, and this is the paper's strongest material: 1. Cost is stable, location is not. 49 of 180 route-years fell below 80 percent overlap at 1 km, covering 21 of the 32 "robust" links. Pair 17-24 hit zero in 2020. 2. Vegetation weighting barely changes direction (43 of 45 links keep their sign) but strongly changes magnitude (median absolute change 3.6 percent at veg-low, 9.3 at veg-high). 3. The ranking depends on the link graph. At $k = 2/3/4/5$ you get 31/45/61/77 edges, only $k = 3$ has a bridge, and the top links change at $k = 4$ and 5. TRAP: do not present a single ranking as if the graph construction were given.]

3.5 Independent plausibility assessment

[TODO: Outcome of §2.11. Report it even if weak. FACTS: early 2010-2012 = 413 thinned 10 km cells, late 2013-2016 = 450, Jaccard 0.239. Late cells sit in slightly higher modelled resistance (medians 2.28 vs 2.13, Cliff's delta 0.11). More late cells fall inside the fixed cores (67.8 vs 45.8 percent). Almost none fall near a priority path (0 of 413, 3 of 450). TRAP: this is not independent. The same archive defined the cores, so the core result is circular by construction. And there are no records after 2016, so 2020 and 2024 have no check at all. Say both.]

3.6 Monitoring priorities

[TODO: H4. No result. The cell-level index and the W1–W3 weight sets were never built. Simplest: say that here, and that the monitoring output is the link-level robust / uncertain split instead. One short subsection explaining what you planned, what you built, and why the gap, is a perfectly good thing to have in a capstone.]

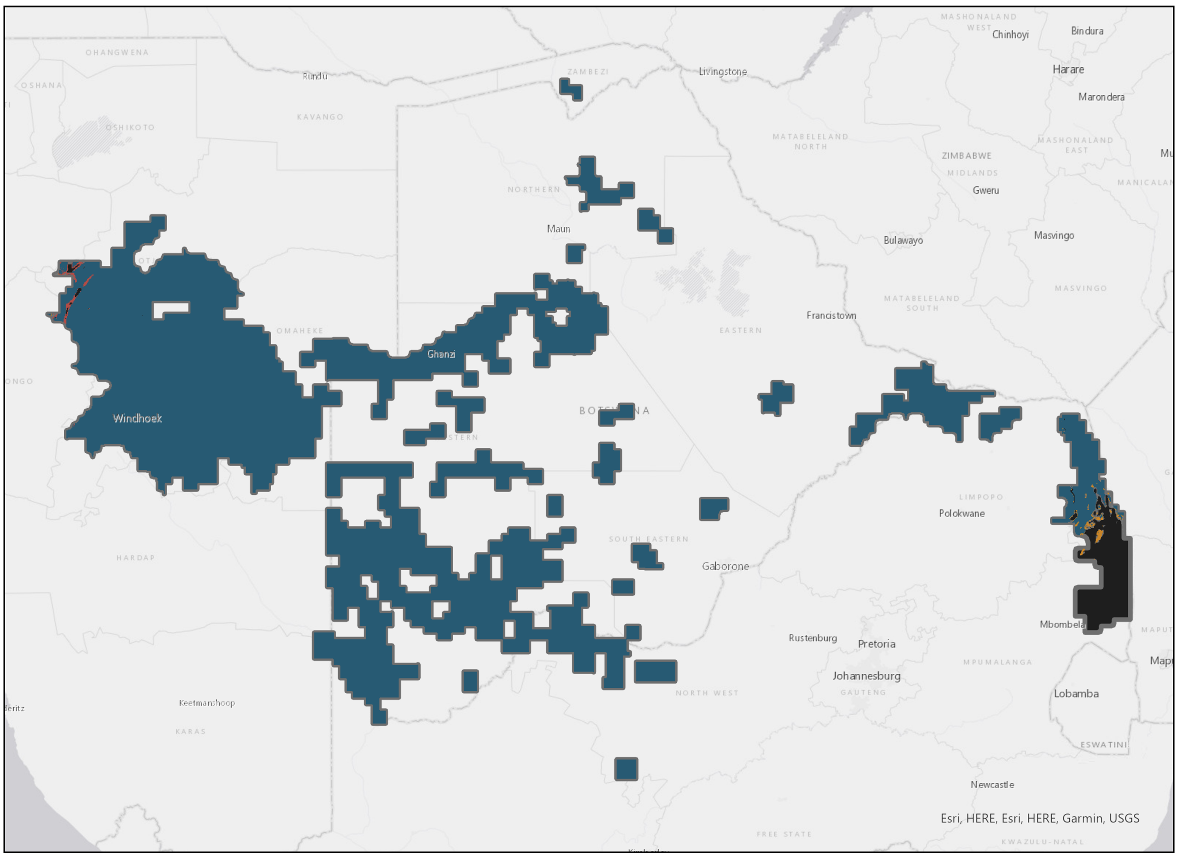

Figure 1: [TODO: Caption describes a superseded figure. The rebuilt version is fig2_current_change_outside_cores in resistances/figures, which shows percent change 2012 to 2024 outside the cores. Rewrite for that, and keep the line that these are modelled structural flow, not validated corridors.]

Table 1: Covariates, native resolution, source years, and static/dynamic treatment.

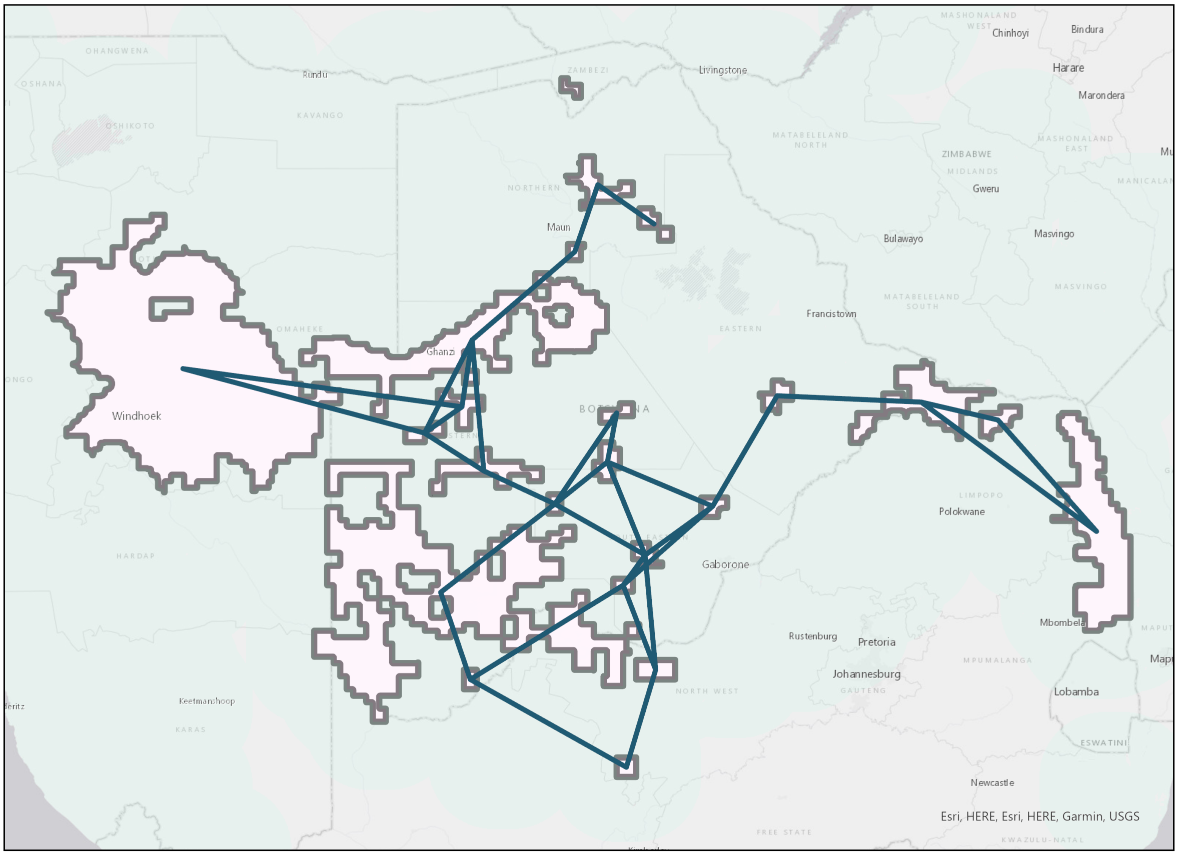

Figure 2: [TODO: Superseded. This file draws straight centroid-to-centroid lines, not routes. Use fig3_priority_links_least_cost_paths in resistances/figures, which draws the real least-cost paths and separates the 32 primary from the 13 uncertainty links.]

Table 2: [TODO: There are 32 priority links, not 20. No country attribution exists and no agreement score exists. Source is table_primary_priority_links in the results package; usable columns are rank, betweenness, link type, straight-line distance, the three robustness flags, and effective resistance per year.]

Table 3: [TODO: Rewrite for the outside-core domain and drop "predeclared null". Source is outside_core_current_pa_curve in resistances/reports.]

Table 4: [TODO: No composite score and no alternative weightings exist. If the decision table replaces the index, retitle to the fields kept separate. Otherwise cut this float.]

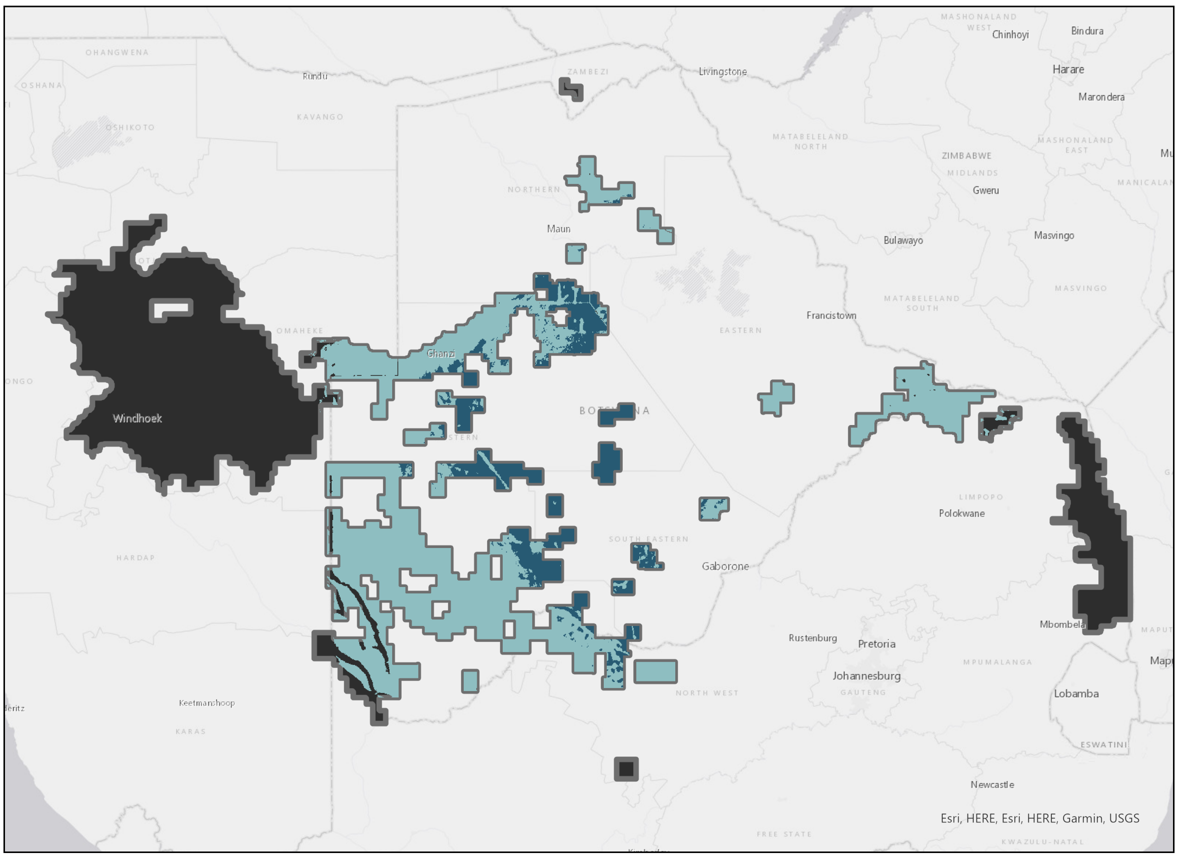

Figure 3: [TODO: Caption and figure do not match. This file is high current concentration in 2024, not ensemble agreement, and no agreement surface exists. Either point this at fig1_current_outside_cores_2024 and retitle, or cut the float.]

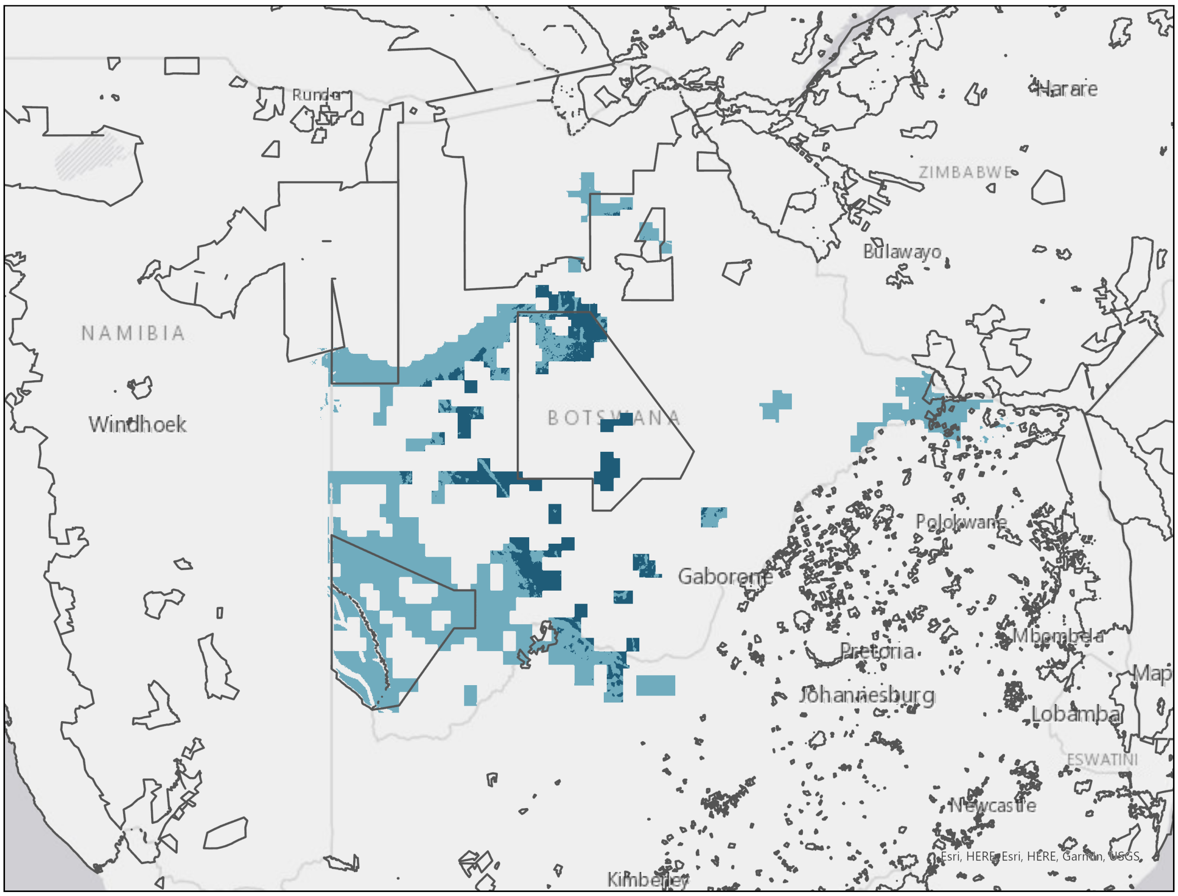

Figure 4: [TODO: Replace with the coverage-versus-percentile curve, outside_core_pa_coverage_curve in resistances/figures. It carries the actual finding: coverage dips to a minimum near the 80th percentile and crosses background near the 93rd.]

4 Discussion

What the results mean, at the level the evidence supports

[TODO: First paragraph restates the headline finding in one sentence and immediately bounds it. Resist writing "cheetah movement" anywhere in this section.]

4.1 Where permeability has declined, and what co-occurred

[TODO: H1. Spatial co-occurrence, never causation (CAU-014). NOTE: the heading says "declined". Permeability did not broadly decline; resistance fell slightly on balance and the direction is mixed. Retitle, or the section argues against its own numbers. Per-country heterogeneity cannot be reported for connectivity, only for covariate change, so either build the attribution or say why it is absent.]

4.2 The protection gap as a governance question

[TODO: H2. Governance framing. THE FINDING IS SHARPER THAN "unprotected": the most concentrated flow is inside parks, and the moderately-high connecting landscape around them is the least protected part of the model domain, bottoming out near the 80th percentile of current. SAY: connectivity for this species depends on land under other management. That is a statement about where effort has to operate, not evidence that protected areas failed. Keep the confound that parks are often sited on land of low agricultural value. TRAP: do not write "connectivity is unprotected". The tail is well protected.]

4.3 Robust versus model-sensitive linkages

[TODO: H3. This is the paper's most transferable contribution. Robust locations support action; model-sensitive locations define the survey agenda. Note the caveat that scenarios sharing input layers can agree artificially.]

4.4 From uncertainty to a survey design

[TODO: H4. There is no monitoring index (CAU-020). Write this about the link-level robust / uncertain split instead. SAY: what a partner organisation would actually do with 13 uncertainty links, and that accessibility and cost will reorder any list in practice. STRONGEST POINT AVAILABLE: pair 17-24 is simultaneously the only bridge in the k=3 graph and the least spatially stable link in the set, zero route overlap in 2020. That is the clearest example of a place where a survey would change the answer.]

4.5 Comparison with regional connectivity work

[TODO: Position against the elephant and multispecies transfrontier assessments (Brennan et al., 2020; Naidoo et al., 2024) as methodological neighbours, and against the fencing-effects literature (Osipova et al., 2018). Be explicit that those are different species with different movement ecology: the methods transfer, the resistance values do not (CAU-015).]

5 Limitations

Fixed historical cores. [TODO: CAU-002. The single most important limitation. Cores derive from 2010–2016 evidence and do not move; all temporal signal is landscape-side.]

Structural, not functional. [TODO: CAU-001.]

Sampling bias in the occurrence baseline. [TODO: CAU-003. Absence of records is not evidence of absence.]

Resistance is expert-assigned. [TODO: CAU-005. What the scenario ensemble does and does not compensate for.]

Static covariates. [TODO: Roads, livestock, fences and terrain contribute no temporal signal; consequences for interpreting change.]

Mapping completeness varies by country. [TODO: CAU-010.]

Geographic transferability. [TODO: CAU-015.]

Scope. [TODO: CAU-018. Climate, conflict and occupancy are deliberately out.]

Rainfall is not in the model. [TODO: CHIRPS was never acquired. Vegetation carries most of the temporal signal and the vegetation term is effectively bare-cover fraction, which in a semi-arid system tracks rainfall. So some of what is reported as landscape change may be interannual rainfall. This is a real limitation, not a footnote.]

The vegetation term is narrower than it sounds. [TODO: The transform reduces to 1 + 0.09 x bare whenever the three VCF fractions sum to 100. Swapping tree for non-tree cover at fixed bare cover changes nothing. Describe it as a bare-cover proxy, not a model of vegetation structure.]

Only one core definition, one resolution. [TODO: The planned three core definitions and the 2 km and 5 km resolution checks were never run, so the claim in §2.2 that conclusions are not resolution artefacts is currently unsupported. Either run them or move that claim here.]

Current is reported outside the cores. [TODO: In a pairwise formulation the cores inject and drain the current, so they carry the top decile by construction. Masking them cleans up what is reported but does not remove focal-region influence from the underlying solve. Also note the resistance graph has 34 connected components.]

Route location is less certain than route cost. [TODO: 49 of 180 route-years fell below 80 percent overlap at 1 km, including 21 of the 32 priority links. A link can be confidently cheap and still not have a confidently located route.]

Input series are not perfectly comparable. [TODO: VIIRS switches from V21 to V22 at the 2024 snapshot, breaking the pin-one-version rule. GHSL 2010 stands in for 2012, exactly at the two-year tolerance. Lights carry the smallest weight, which limits the damage, but say it.]

6 Conclusion

[TODO: Three or four sentences. What changed, where the robust priorities are and how protected, the monitoring recommendation, the claim boundary. No new information. TRAPS, all three easy to walk into here: Not "connectivity declined". It barely moved, and mixed. Not "connectivity is unprotected". The concentrated tail is well protected; the moderately-high connecting landscape is not. Not a monitoring index. The output is 32 robust links and 13 that need survey.]

Data availability

[TODO: CAU-012, CAU-013. Derived outputs and all code are archived at [TODO: OSF/Zenodo DOI]. Raw protected-area data are not redistributed and must be obtained from Protected Planet under its terms. Occurrence data are shared only at the resolution permitted by the source archive.]

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[TODO: Department, any data providers, any partner organisation that reviewed the monitoring priorities.]

Conflicts of interest

The author declares no conflicts of interest.

References

Adriaensen, F., Chardon, J. P., De Blust, G., Swinnen, E., Villalba, S., Gulinck, H., & Matthysen, E. (2003). The application of 'least-cost' modelling as a functional landscape model. *Landscape and Urban Planning*, *64*(4), 233–247. https://doi.org/10.1016/S0169-2046(02)00242-6

Anantharaman, R., Hall, K., Shah, V. B., & Edelman, A. (2020). Circuitscape in Julia: High performance connectivity modelling to support conservation decisions. *Proceedings of the JuliaCon Conferences*, *1*(1), 58. https://doi.org/10.21105/jcon.00058

Beier, P., Majka, D. R., & Spencer, W. D. (2008). Forks in the road: Choices in procedures for designing wildland linkages. *Conservation Biology*, *22*(4), 836–851. https://doi.org/10.1111/j.1523-1739.2008.00942.x

Brennan, A., Beytell, P., Aschenborn, O., Du Preez, P., Funston, P. J., Hanssen, L., Kilian, J. W., Stuart-Hill, G., Taylor, R. D., & Naidoo, R. (2020). Characterizing multispecies connectivity across a transfrontier conservation landscape. *Journal of Applied Ecology*, *57*(9), 1700–1710. https://doi.org/10.1111/1365-2664.13716

Buk, K. G., van der Merwe, V. C., Marnewick, K., & Funston, P. J. (2018). Conservation of severely fragmented populations: Lessons from the transformation of uncoordinated reintroductions of cheetahs (*Acinonyx jubatus*) into a managed metapopulation with self-sustained growth. *Biodiversity and Conservation*, *27*(13), 3393–3423. https://doi.org/10.1007/s10531-018-1606-y

Keeley, A. T. H., Beier, P., & Jenness, J. S. (2021). Connectivity metrics for conservation planning and monitoring. *Biological Conservation*, *255*, 109008. https://doi.org/10.1016/j.biocon.2021.109008

Landau, V. A., Shah, V. B., Anantharaman, R., & Hall, K. R. (2021). Omniscape.jl: Software to compute omnidirectional landscape connectivity. *Journal of Open Source Software*, *6*(57), 2829. https://doi.org/10.21105/joss.02829

Marker, L. L., Dickman, A. J., Mills, M. G. L., Jeo, R. M., & Macdonald, D. W. (2008). Spatial ecology of cheetahs on north-central namibian farmlands. *Journal of Zoology*, *274*(3), 226–238. https://doi.org/10.1111/j.1469-7998.2007.00375.x

McRae, B. H., Dickson, B. G., Keitt, T. H., & Shah, V. B. (2008). Using circuit theory to model connectivity in ecology, evolution, and conservation. *Ecology*, *89*(10), 2712–2724. https://doi.org/10.1890/07-1861.1

Naidoo, R., Beytell, P., Brennan, A., Carter, J., Carter, K. D., Chamaillé-Jammes, S., Chilambe, B., Hoare, R., Iiyambo, N., Jooste, D., Karidozo, M., Kilian, J. W., Madhlamoto, D., Madiri, T. H., McCulloch, G., Monks, N., Mudimba, I., Ngwenya, N., Nyambe, N., … Songhurst, A. (2024). Landscape connectivity for african elephants in the world's largest transfrontier conservation area: A collaborative, multi-scalar assessment. *Journal of Applied Ecology*, *61*(10), 2483–2496. https://doi.org/10.1111/1365-2664.14746

Osipova, L., Okello, M. M., Njumbi, S. J., Ngene, S., Western, D., Hayward, M. W., & Balkenhol, N. (2018). Fencing solves human-wildlife conflict locally but shifts problems elsewhere: A case study using functional connectivity modelling of the african elephant. *Journal of Applied Ecology*, *55*(6), 2673–2684. https://doi.org/10.1111/1365-2664.13246

Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J. J., Schröder, B., Thuiller, W., Warton, D. I., Wintle, B. A., Hartig, F., & Dormann, C. F. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, *40*(8), 913–929. https://doi.org/10.1111/ecog.02881

Saura, S., & Pascual-Hortal, L. (2007). A new habitat availability index to integrate connectivity in landscape conservation planning: Comparison with existing indices and application to a case study. *Landscape and Urban Planning*, *83*(2-3), 91–103. https://doi.org/10.1016/j.landurbplan.2007.03.005

Sousa, L. L., Kotze, R., Fanikiso, L., Young-Overton, K., Overton, J., Stevens, X., Lines, R., Wijers, M., Kesch, K., Bauer, D. T., Seymour-Smith, J., Sibanda, A., Parry, R., Fitt, M., Burger, K., Seyoka, S., McNutt, J. W., Stein, A. B., Pelayo-Malet, V., . . . Loveridge, A. J. (2026). Multispecies transboundary landscape connectivity in the KAZA TFCA. *Scientific Reports*, *16*(1), 24330. https://doi.org/10.1038/s41598-026-58440-y

Zeller, K. A., Jennings, M. K., Vickers, T. W., Ernest, H. B., Cushman, S. A., & Boyce, W. M. (2018). Are all data types and connectivity models created equal? validating common connectivity approaches with dispersal data. *Diversity and Distributions*, *24*(7), 868–879. https://doi.org/10.1111/ddi.12742

Zeller, K. A., McGarigal, K., & Whiteley, A. R. (2012). Estimating landscape resistance to movement: A review. *Landscape Ecology*, *27*(6), 777–797. https://doi.org/10.1007/s10980-012-9737-0